

Characteristic-Time Grid Study

Mass at t^* versus Grid Size

18 August 2026 · extended to 224 x 224, 26 August 2026

Parameters

Held fixed across every grid:

Microtubules	$N = 16$
Advective velocity	$v = 10^4$
Switch rates	$a = b = w = 100$
Dimensionless solution time	$T = 1$
Domain radius	$R = 1.0$
Diffusion coefficient	$D = 1.0$
Microtubule extraction width	$d_{\text{tube}} = 0.0$
Initial condition	centred

Results and wall time

Grid	t^*	m^*	m	Device	Wall time
48 x 48	0.052357	0.670035	-0.369056	CPU	3.8 min
64 x 64	0.055390	0.546092	-0.527038	CPU	19.3 min
80 x 80	0.057689	0.450505	-0.666989	CPU	1.23 hr
96 x 96	0.059530	0.377297	-0.790103	CPU	3.68 hr
112 x 112	0.061058	0.320804	-0.898529	CPU	9.32 hr
128 x 128	0.062355	0.276652	-0.994425	CPU	21.31 hr
144 x 144	0.063476	0.241655	-1.079686	CPU	43.68 hr
160 x 160	0.064458	0.213518	-1.155910	CPU	82.36 hr
176 x 176	0.065327	0.190594	-1.224422	GPU	22.48 hr
192 x 192	0.066104	0.171680	-1.286315	GPU	30.32 hr
208 x 208	0.066804	0.155899	-1.342492	GPU	35.26 hr
224 x 224	0.067439	0.142591	-1.393708	GPU	54.01 hr

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